WQF7004
Data Analytics

Introduction to Databases and CRISP-DM

Learning Outcomes

- In this module, you will learn about databases and the process of CRISP-DM.

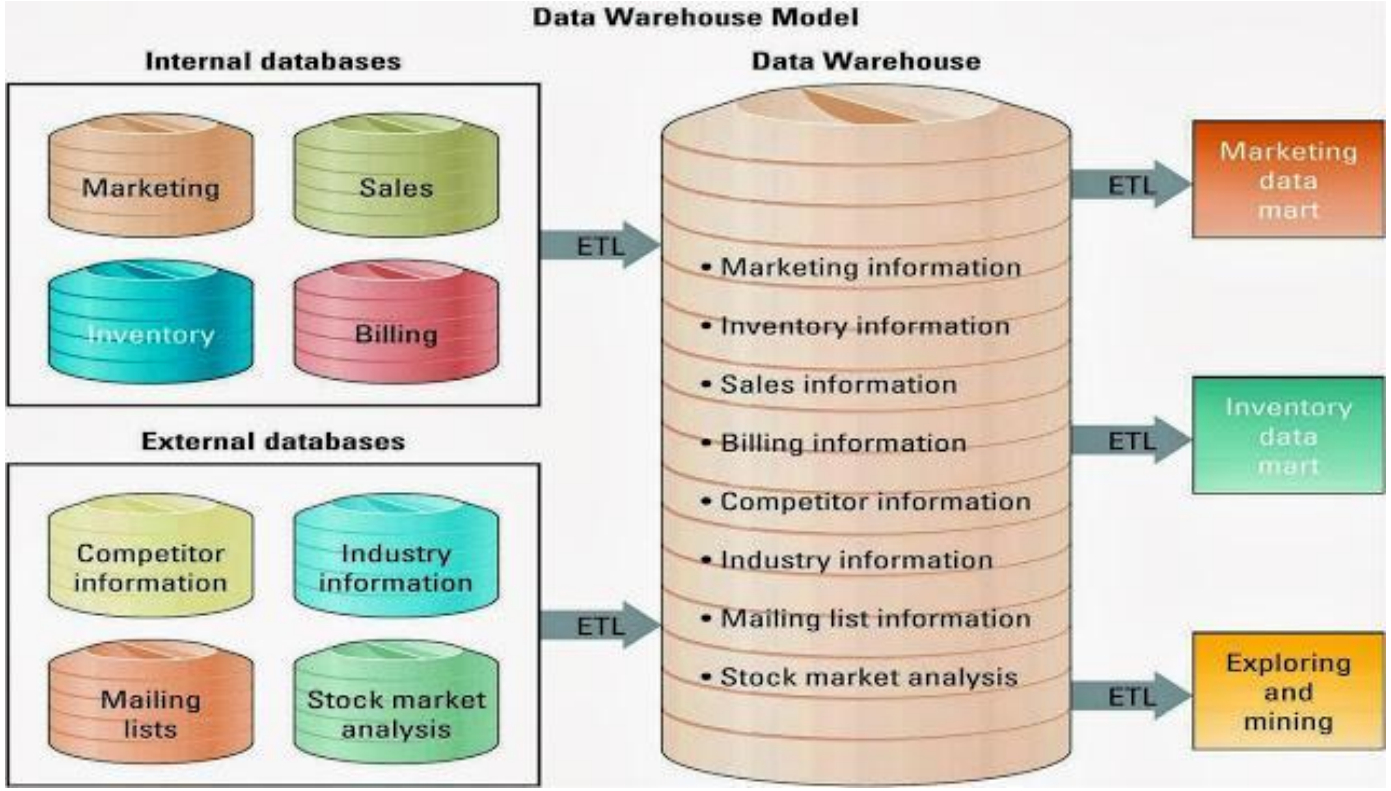
Topic Title	Topic Objective
Databases	Differentiate between data warehouse, database and data mart.
Data Science Methodology	Describe the process of CRISP-DM

Contents

- Type of data storage
- Data Communication
- CRISP-DM Process
- Summary
- Key Terms
- References

Data Storage

- Data stored in different form as shown below for different purposes.
  - » data warehouse
  - » database
  - » data mart



Data Storage

Data Warehouse

- A logical collection of information gather from many operational databases. It supports business analytics activities and Decision-making task

Database

- An organized grouping of information within a specific structure that needs to be retrieved frequently.

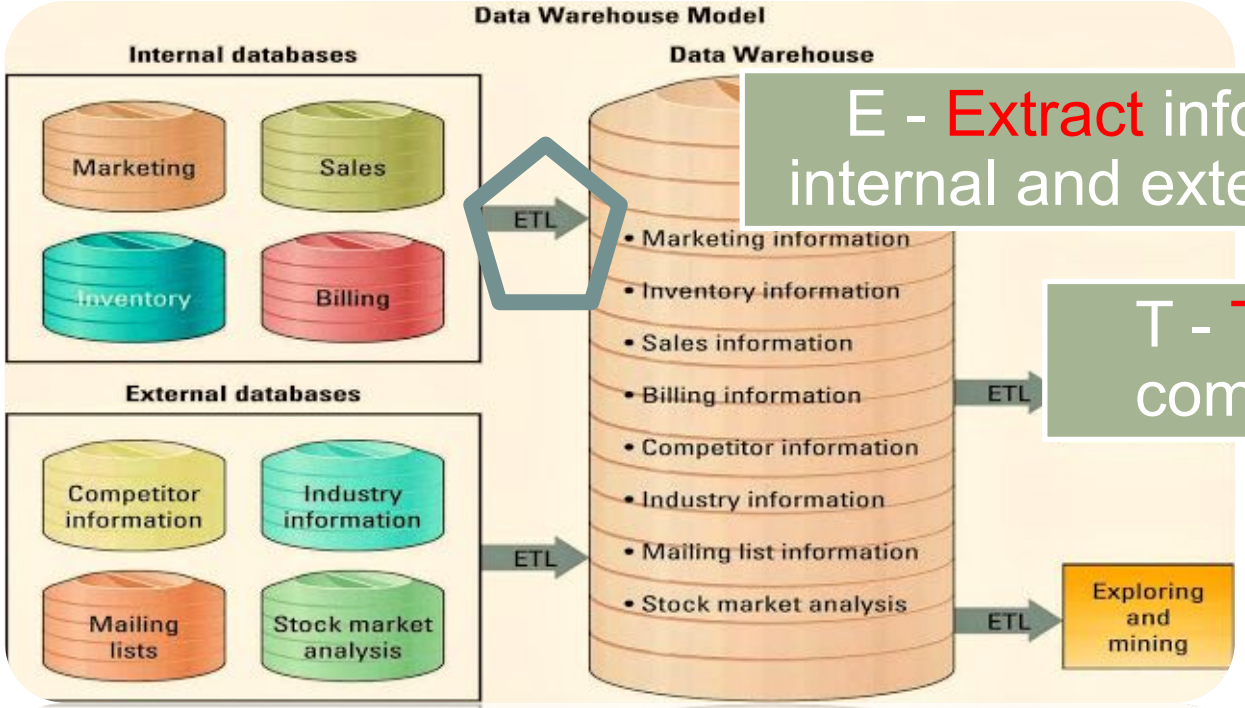
Data Mart

- Subsets of data warehouses that is highly focused and isolated for a specific population of users

Database vs Data Warehouse

	Database	Data Warehouse
Purpose	For data retrieval, updating and management	For data analysis and decision making
Systems / Application	OLTP (OnLine Transaction Processing System)	Analytical Software. i.e. data mining tools, reporting tools and OLAP (OnLine Analytical Processing) tools
Format	• Normalised • Relational database • Lowest level of granularity (i.e. individual transactions)	• Denormalised and integrated • Multi-dimensional arrays or relational format • Subject-oriented • Granularity level depends on subject
Time Frame	Current / Real time	Historical

Data Communication



E - Extract information from internal and external databases

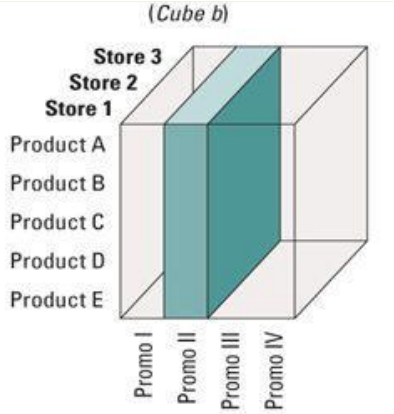
T - Transform the information using common set of enterprise definition

L - Load the information into a data warehouse

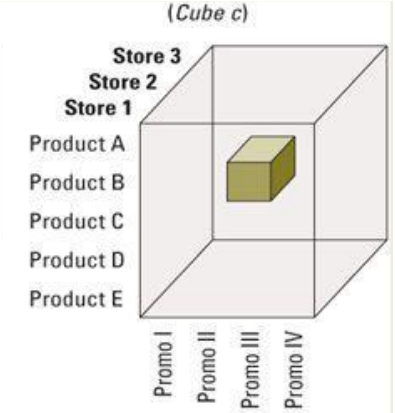
Data Storage

Multidimensional Information

- Store information (layers),
- product information (rows),
- promotion information (columns)

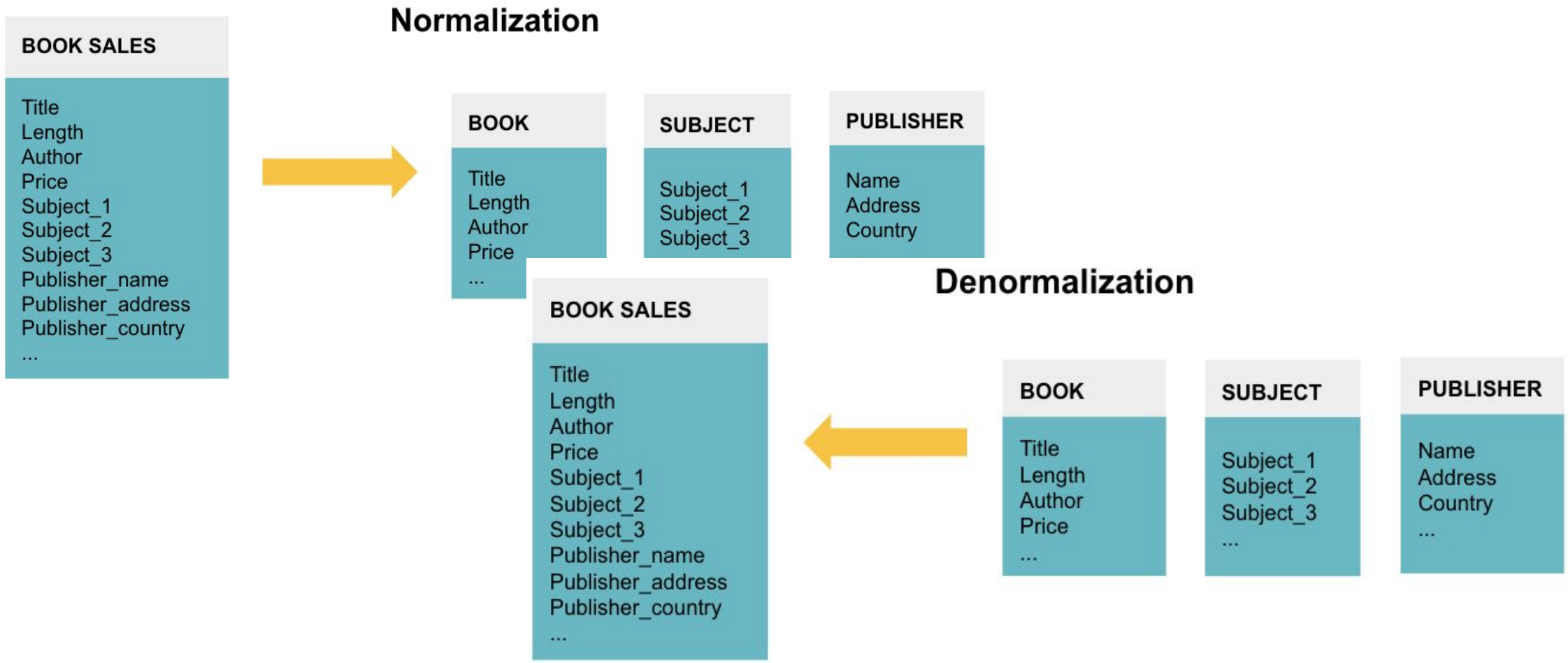


- A slice of information displaying Promotion II for all products at all stores.



- A slice of information displaying Promotion II for Product B at Store 2.

Normalised Vs Denormalised data



Relational database

- A relational database organizes data into tables which can be linked—or related—based on data common to each.
- Table 1 contains information about pet owners. Table II contains information about pets
- The tables are related by the column Owner_ID.
- By relating tables to one another, we can reduce data redundancy and improve database performance.

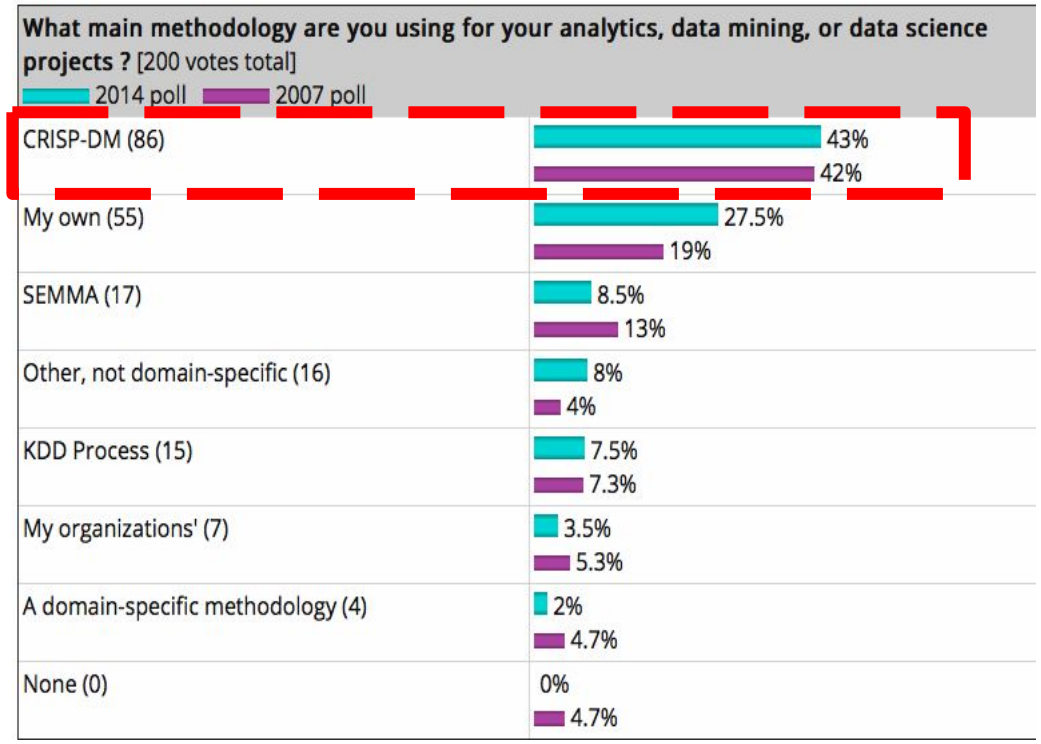
Owner_ID	Owner Name
1	Jim
2	Joan

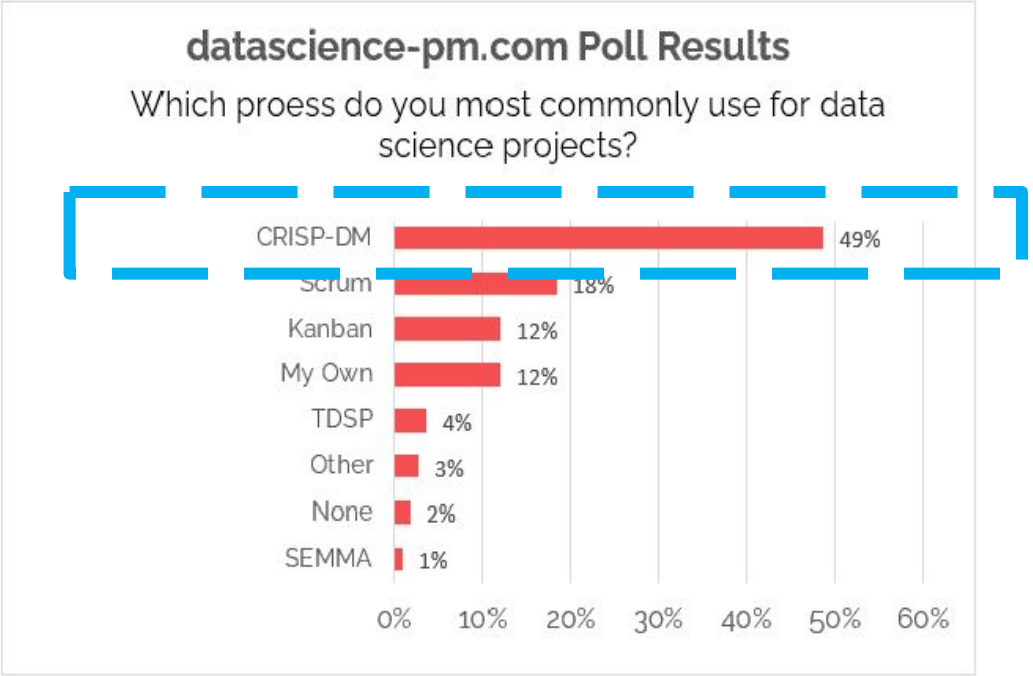
Pet_ID	Pet_Name	Owner_ID
1	Fifi	2
2	Butch	1
3	Clover	2
4	Animal	1
5	Tank	1

Dataset

- A data set is a subset of a database or a data warehouse.
- It is usually denormalized so that only one table is used.
- The creation of a data set may contain several steps, including appending or combining tables from source database tables, or simplifying some data expressions.
- Data sets may be made up of a representative sample of a larger set of data, or they may contain all observations relevant to a specific group.

Data Science Methodology





Kdnuggets 2014 polls

CRISP-DM

CRoss-Industry Standard Process for Data Mining

- An open **standard process** model that describes common approaches used by data mining experts.
- The data mining process must be **reliable and repeatable by people with little data mining background.**
- **widely-used** analytics model.

CRISP-DM History

Initiative launched in late 1996 by three "veterans" of data mining market.

Daimler Chrysler (then Daimler-Benz), SPSS (then ISL) , NCR

Developed and refined through series of workshops (from 1997-1999).

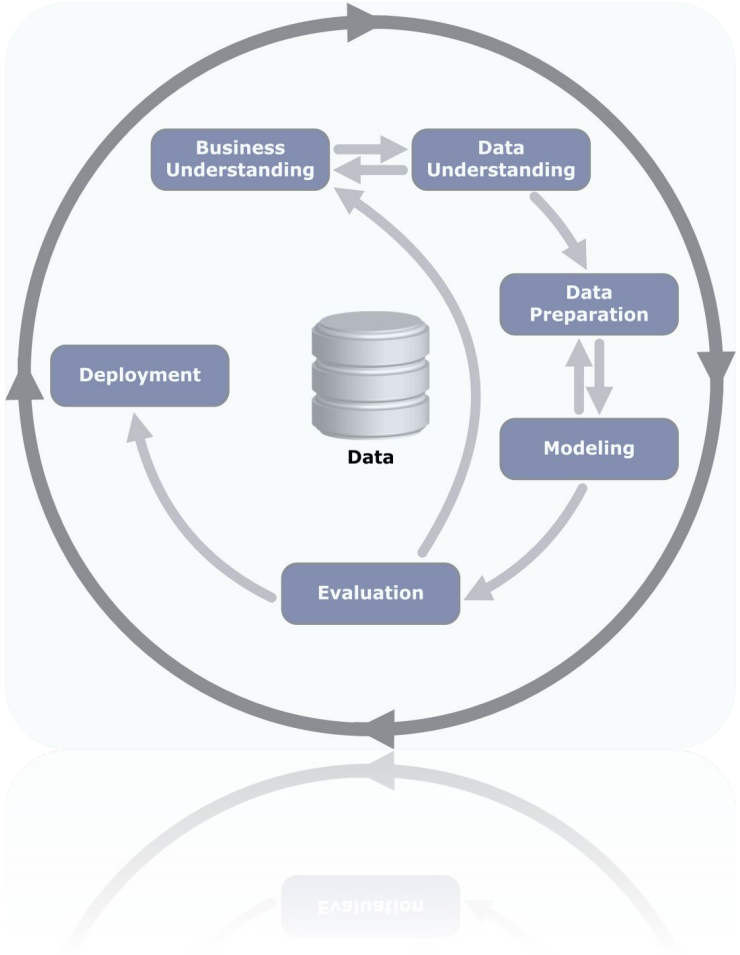
Published CRISP-DM 1.0 (1999)

Over 200 members of the CRISP-DM SIG worldwide.

CRISP-DM

- Non-proprietary
- Application/Industry neutral
- Tool neutral
- Focus on business issues and technical analysis
- Framework for guidance
- Experience base templates for Analysis

CRISP-DM



- The life cycle of a data mining project consists of six phases.
- Not rigid process □ moving back and forth between each phase determines which phases has to be performed next.

Overview of CRISP-DM

Business Understanding

Determine Business Objectives
Background
Business Objectives
Business Success Criteria

Situation Assessment
Inventory of Resources
Requirements, Assumptions, and Constraints
Risks and Contingencies
Terminology
Costs and Benefits

Determine Data Mining Goal
Data Mining Goals
Data Mining Success Criteria

Produce Project Plan
Project Plan
Initial Asessment of Tools and Techniques

Data Understanding

Collect Initial Data
Initial Data Collection Report

Describe Data
Data Description Report

Explore Data
Data Exploration Report

Verify Data Quality
Data Quality Report

Data Preparation

Data Set
Data Set Description

Select Data
Rationale for Inclusion / Exclusion

Clean Data
Data Cleaning Report

Construct Data
Derived Attributes
Generated Records

Integrate Data
Merged Data

Format Data
Reformatted Data

Modeling

Select Modeling Technique
Modeling Technique
Modeling Assumptions

Generate Test Design
Test Design

Build Model
Parameter Settings
Models
Model Description

Assess Model
Model Assessment
Revised Parameter Settings

Evaluation

Evaluate Results
Assessment of Data Mining Results w.r.t. Business Success Criteria
Approved Models

Review Process
Review of Process

Determine Next Steps
List of Possible Actions
Decision

Deployment

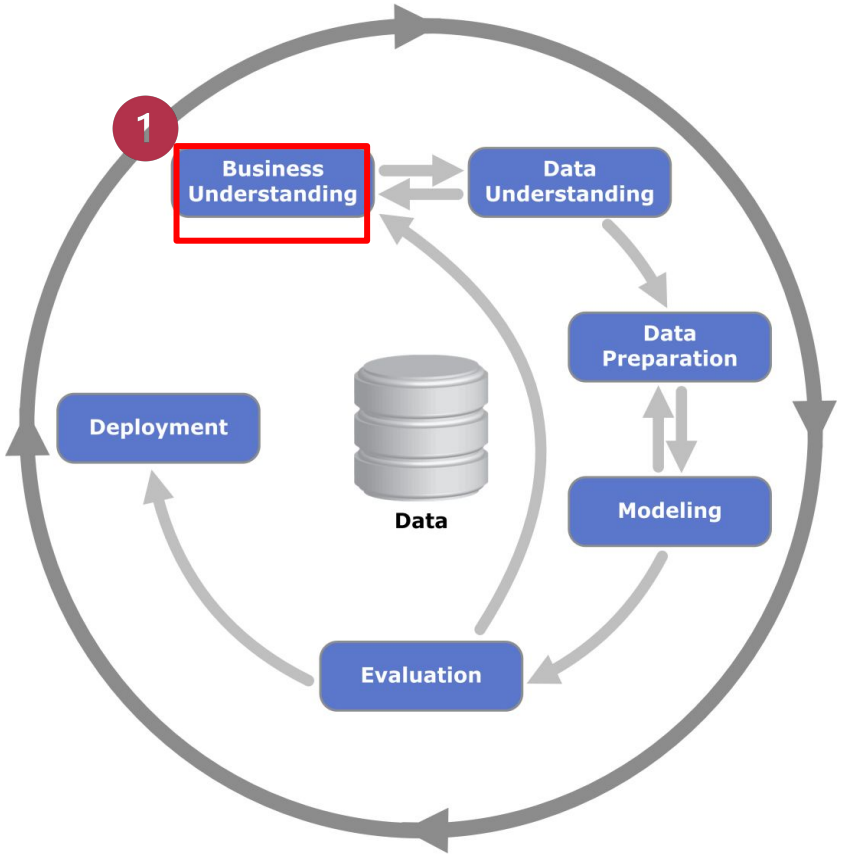
Plan Deployment
Deployment Plan

Plan Monitoring and Maintenance
Monitoring and Maintenance Plan

Produce Final Report
Final Report
Final Presentation

Review Project
Experience Documentation

Phase 1 - Business Understanding



4 TASKS

1. Determine business objective
1. Assess situation
1. Determine data mining goals
1. Produce project plan

Phase 1 - Business Understanding

Determine business objectives

- Thoroughly understand, from a business perspective, what the client really wants to accomplish.
- Uncover important factors, at the beginning, that can influence the outcome of the project.
- Neglecting this step is to expend a great deal of effort producing the right answers to the wrong questions.

Assess situation

- More detailed fact-finding about all of the resources, constraints, assumptions and other factors that should be considered
- Flesh out the details.

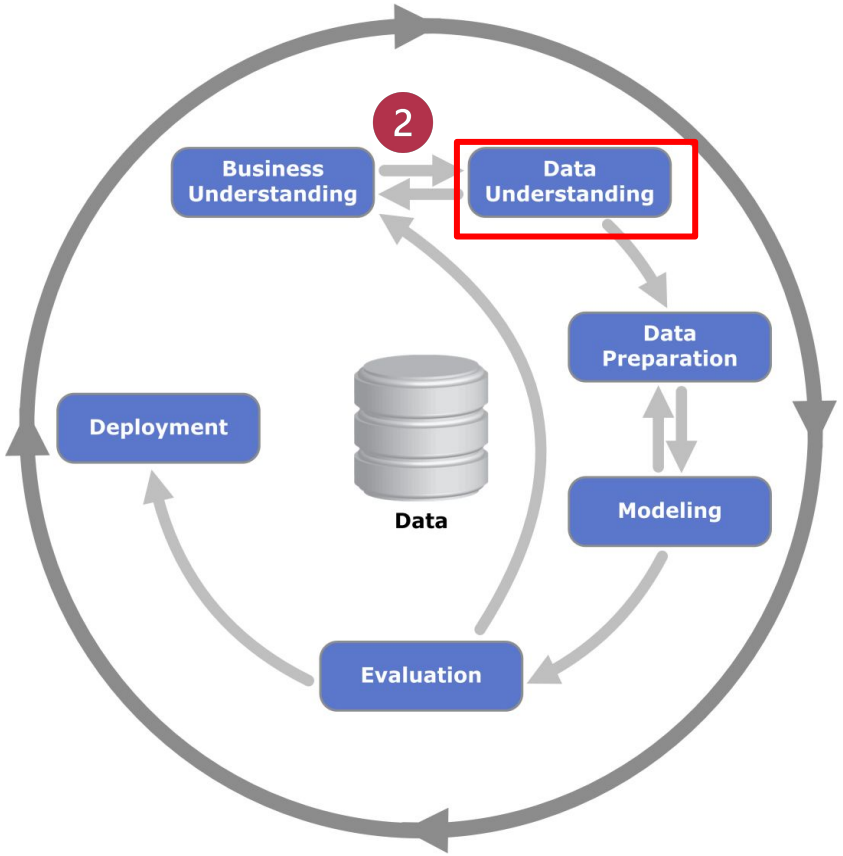
Determine data mining goals

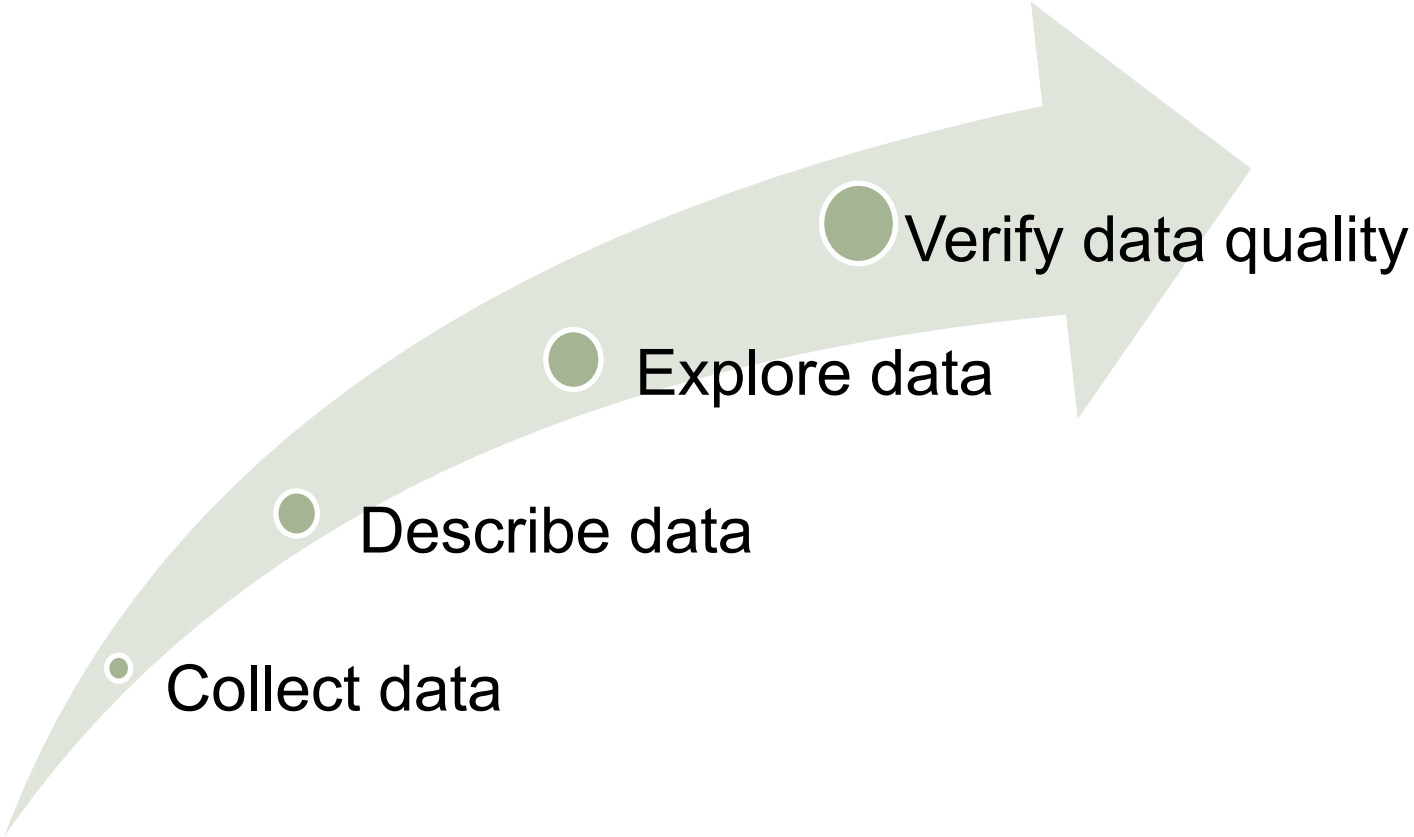
- State business goal
- State data mining goal which include project objectives in technical terms.
- Example
  - Business goal: "Increase catalog sales to existing customers."
  - Data mining goal: "Predict how many widgets a customer will buy, given their purchases over the past three years, demographic information (age, salary, city) and the price of the item."

Produce project plan

- Describe the intended plan for achieving the data mining goals and the business goals.
- the plan should specify the anticipated set of steps to be performed during the rest of the project including an initial selection of tools and techniques.

Phase 2 – Data Understanding





Phase 2 – Data Understanding

1. Collect initial data
   - » acquire the data listed in the project resources
   - » includes data loading if necessary, for data understanding
   - » possibly leads to initial data preparation steps
   - » Notes: if acquiring multiple data sources, integration is an additional issue, either here or in the later data preparation phase

1. Describe data
   - » examine the “gross” or “surface” properties of the acquired data
   - » report on the results

Phase 2 – Data Understanding

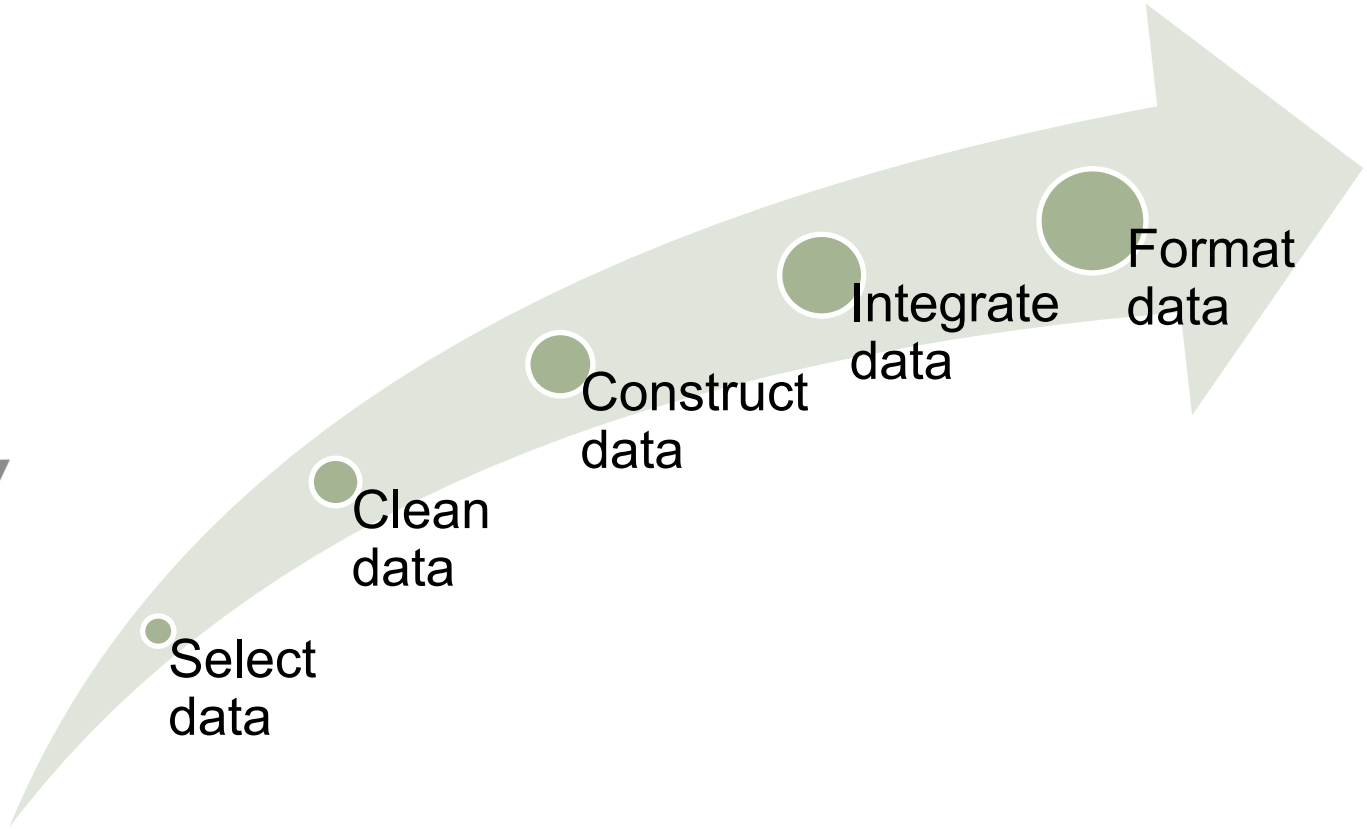
3. Explore data
   - tackles the data mining questions, which can be addressed using querying, visualization and reporting including:
     - distribution of key attributes, results of simple aggregations
     - relations between pairs or small numbers of attributes
     - properties of significant sub-populations, simple statistical analyses
   - may address directly the data mining goals
     - may contribute to or refine the data description and quality reports
     - may feed into the transformation and other data preparation needed

3. Verify data quality
   - examine the quality of the data, addressing questions such as:
     - “Is the data complete?”,
     - Are there missing values in the data?”

Phase 2 – Data Understanding

4. Verify data quality
   - examine the quality of the data, addressing questions such as:
     - “Is the data complete?”,
     - Are there missing values in the data?”

Phase 3 – Data Preparation



Phase 3 – Data Preparation

1. Select data
   - decide on the data to be used for analysis
   - criteria include relevance to the data mining goals, quality and technical constraints such as limits on data volume or data types
   - covers selection of attributes as well as selection of records in a table

1. Clean data
   - raise the data quality to the level required by the selected analysis techniques
   - may involve selection of clean subsets of the data, the insertion of suitable defaults or more ambitious techniques such as the estimation of missing data by modeling.

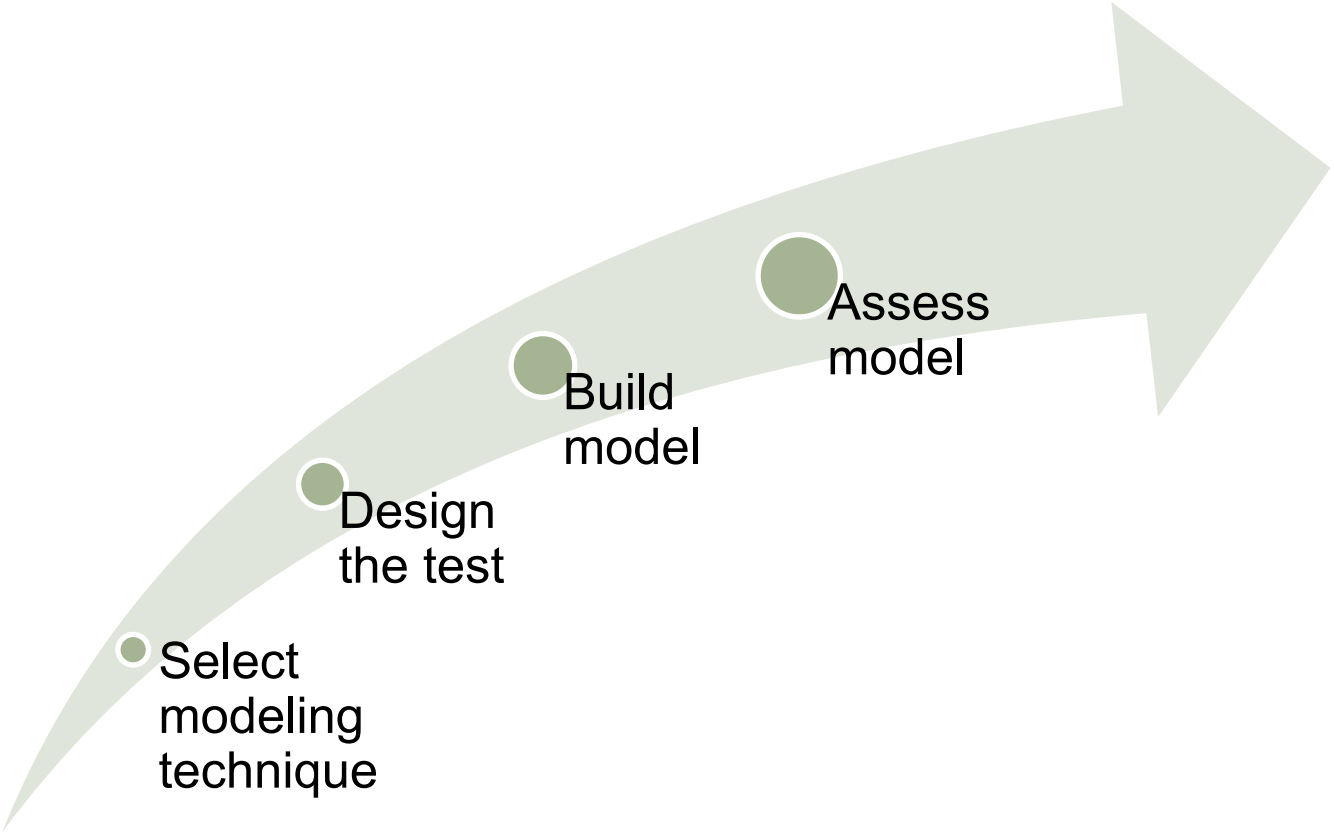
Phase 3 – Data Preparation

3. Construct data
   - constructive data preparation operations such as the production of derived attributes, entire new records or transformed values for existing attributes

3. Integrate data
   - methods whereby information is combined from multiple tables or records to create new records or values

3. Format data
   - formatting transformations refer to primarily syntactic modifications made to the data that do not change its meaning, but might be required by the modeling tool

Phase 4 – Modeling



Phase 4 – Modeling

1. Select the modeling technique (based upon the data mining objective)

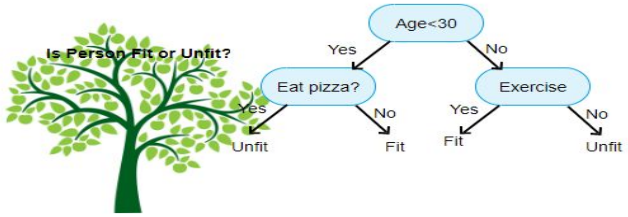
1. Build model (Parameter settings)

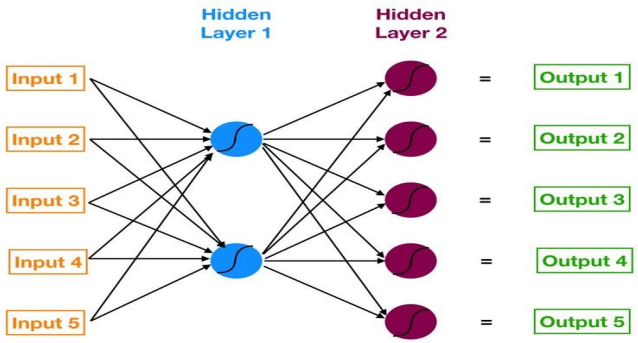
1. Assess model (rank the models)

- Various modeling techniques are selected and applied and their parameters are calibrated to optimal values.
- Some techniques have specific requirements on the form of data.
- Therefore, stepping back to the data preparation phase is often necessary.

Phase 4 – Modeling

Select modelling technique → Generate test design → Build model → Assess model

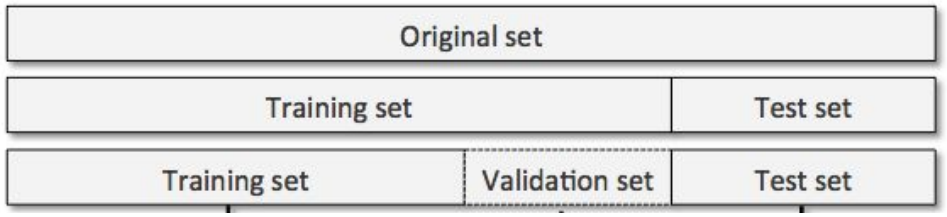




- select the actual modeling technique that is to be used
- i.e. decision tree, neural network
- if multiple techniques are applied, perform this task for each techniques separately

Phase 4 – Modeling

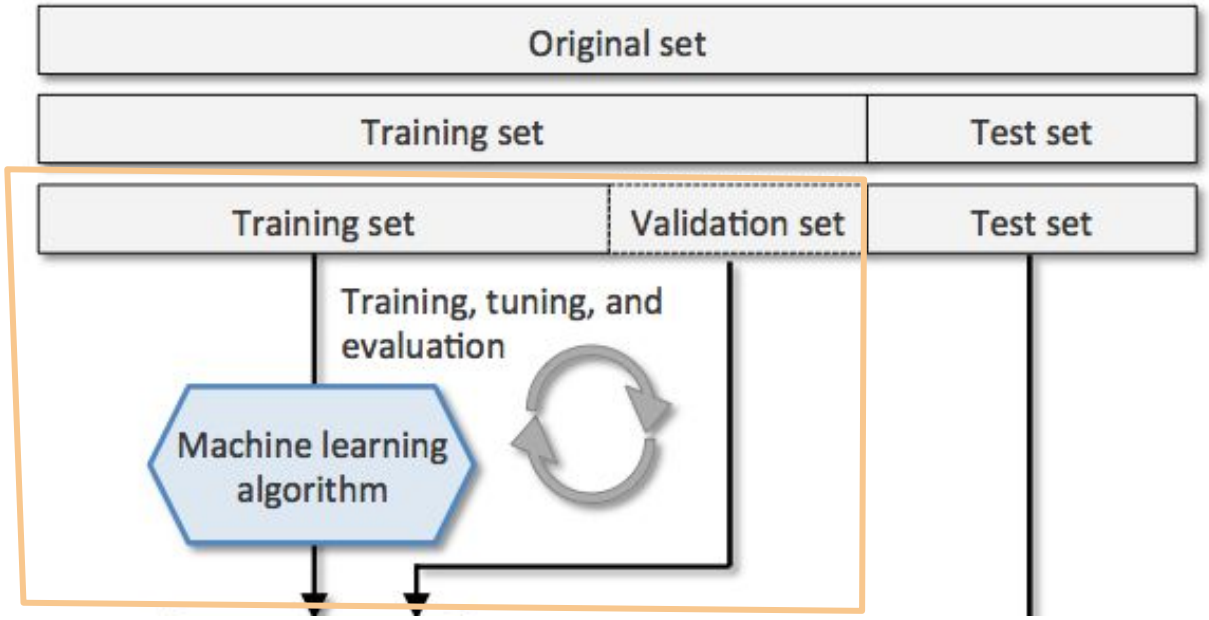
Select modelling technique → **Generate test design** → Build model → Assess model



- before actually building a model, generate a procedure or mechanism to test the model's quality and validity
- i.e. In classification, it is common to use error rates as quality measures for data mining models. Therefore, typically separate the dataset into train and test set, build the model on the train set and estimate its quality on the separate test set

Phase 4 – Modeling

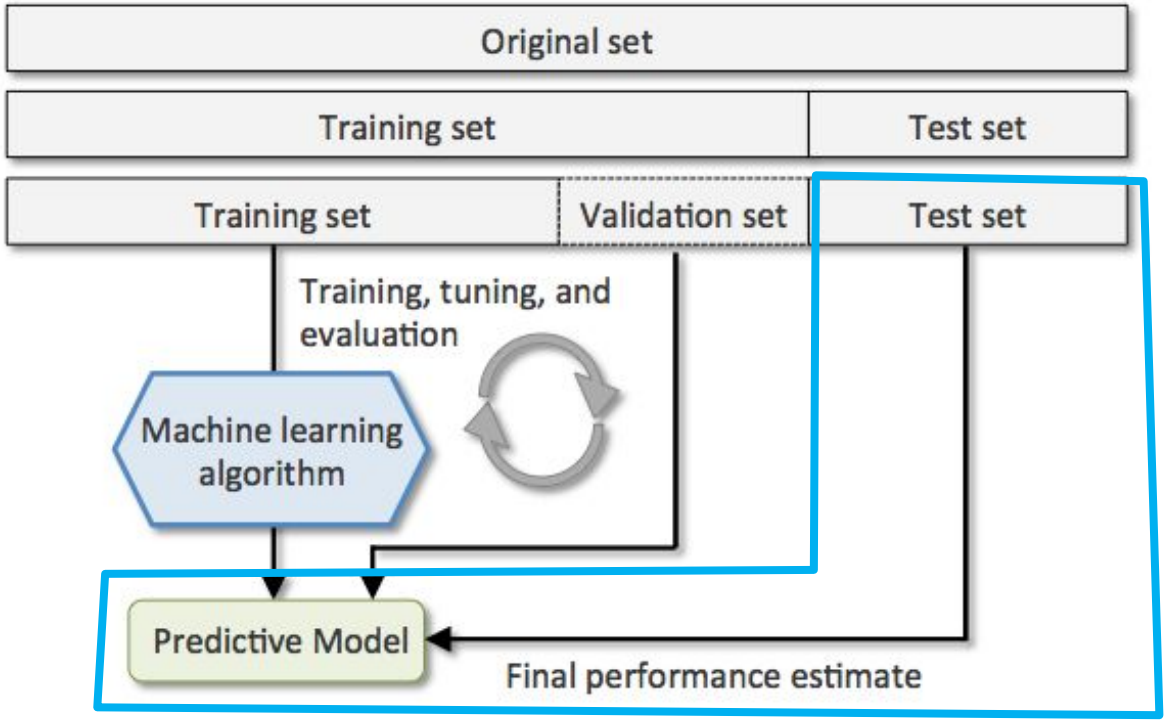
Select modelling technique → Generate test design → **Build model** → Assess model



run the modeling tool on the prepared dataset to create one or more models

Phase 4 – Modeling

Select modelling technique → Generate test design → Build model → **Assess model**



Phase 4 – Modeling

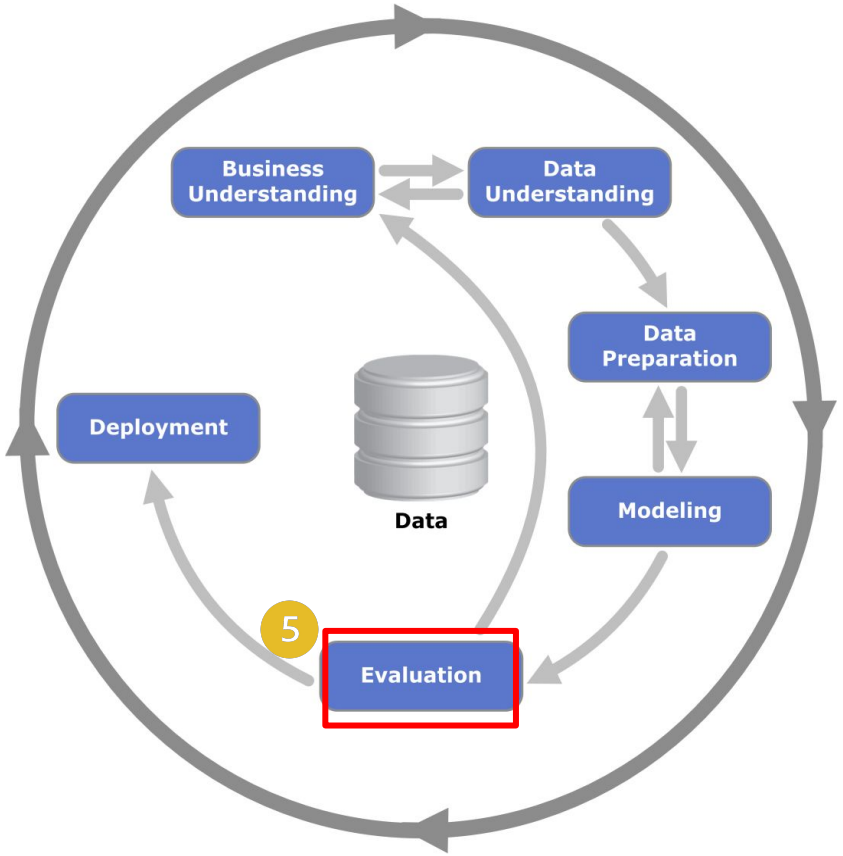
1. Interprets the models according to his domain knowledge, the data mining success criteria and the desired test design

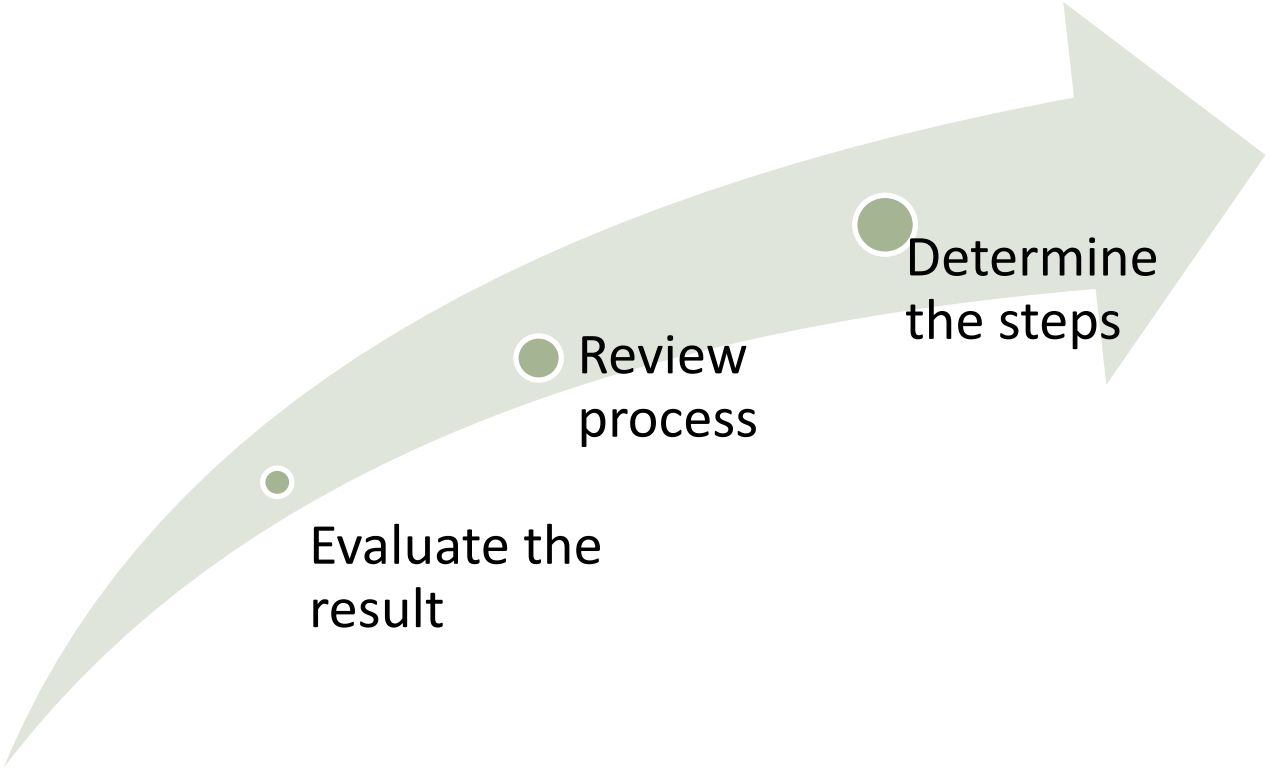
1. Judges the success of the application of modeling and discovery techniques more technically

1. Contacts business analysts and domain experts later in order to discuss the data mining results in the business context

1. Only consider models whereas the evaluation phase also takes into account all other results that were produced in the course of the project

Phase 5 – Evaluation





Phase 5 – Evaluation

1. Evaluate results
   - assesses the degree to which the model meets the business objectives
   - seeks to determine if there is some business reason why this model is deficient
   - test the model(s) on test applications in the real application if time and budget constraints permit
   - also assesses other data mining results generated
   - unveil additional challenges, information or hints for future directions

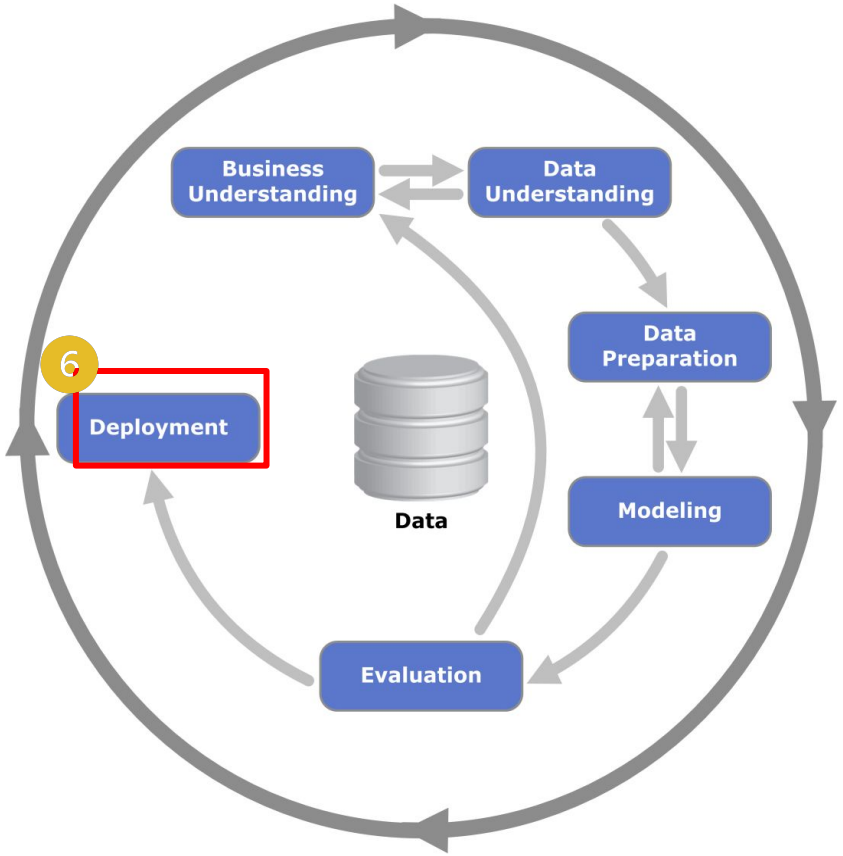
Phase 5 – Evaluation

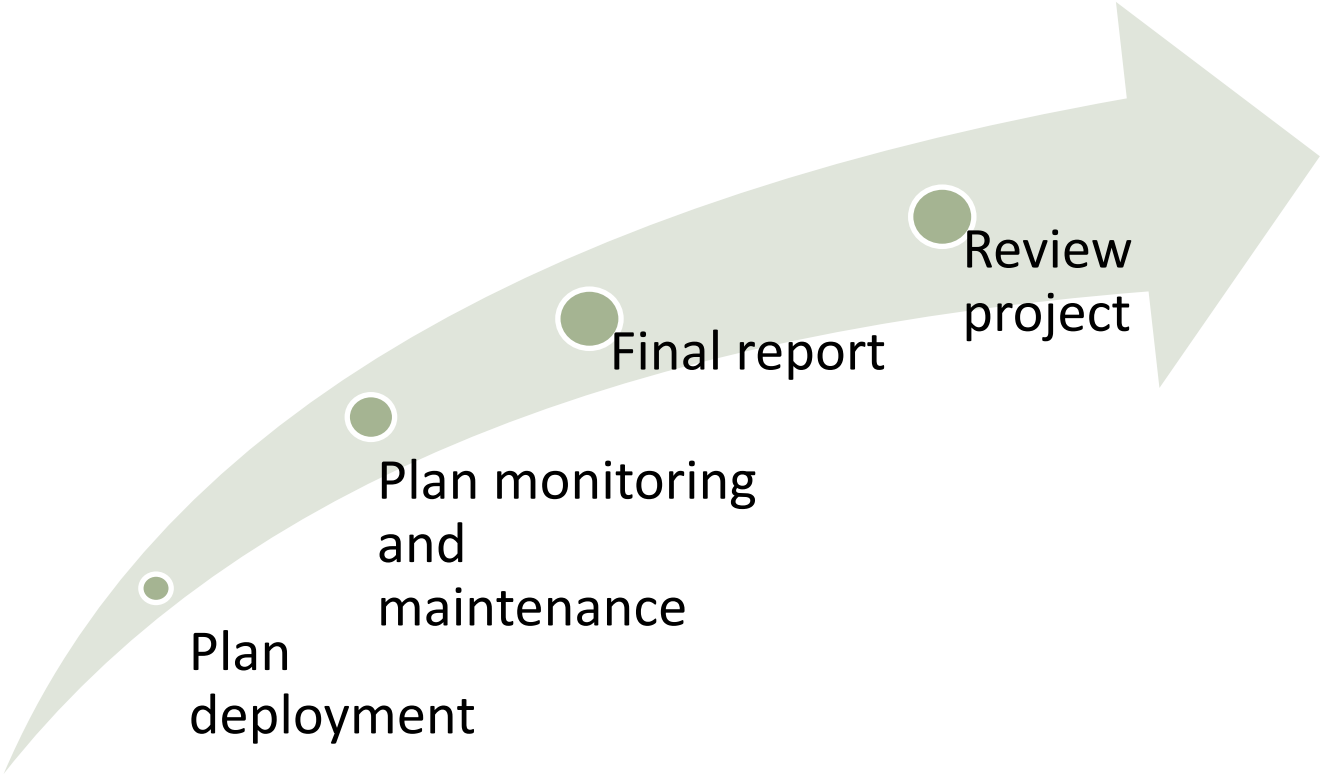
2. Review process
   - do a more thorough review of the data mining engagement in order to determine if there is any important factor or task that has somehow been overlooked
   - review the quality assurance issues
   - i.e. “Did we correctly build the model?”

Phase 5 – Evaluation

3. Determine next steps
   - decides how to proceed at this stage
   - decides whether to finish the project and move on to deployment if appropriate or whether to initiate further iterations or set up new data mining projects
   - include analyses of remaining resources and budget that influences the decisions

Phase 6 – Deployment





Phase 6 – Deployment

1. Plan deployment
   - in order to deploy the data mining result(s) into the business, takes the evaluation results and concludes a strategy for deployment
   - document the procedure for later deployment

1. Plan monitoring and maintenance
   - important if the data mining results become part of the day-to-day business and its environment
   - helps to avoid unnecessarily long periods of incorrect usage of data mining results
   - needs a detailed on monitoring process
   - takes into account the specific type of deployment

Phase 6 – Deployment

1. Produce final report
   - the project leader and his team write up a final report
   - may be only a summary of the project and its experiences
   - may be a final and comprehensive presentation of the data mining result(s)

1. Review project
   - assess what went right and what went wrong, what was done well and what needs to be improved

Summary

- What did I learn in this module?
  - State the difference between data warehouse, database and data mart.
  - Identify and explain the tasks in every CRISP-DM phase

Key Terms

- **Data Warehouse**
  - » A logical collection of information gather from many operational databases. It supports business analytics activities and Decision-making task
- **Database**
  - » An organized grouping of information within a specific structure that needs to be retrieved frequently.
- **Data Mart**
  - » Subsets of data warehouses that is highly focused and isolated for a specific population of users
- **Dataset**
  - » Subsets of data warehouses that is highly focused and isolated for a specific population of users
- **CRIPS-DM**
  - » CRoss-Industry Standard Process for Data Mining
  - » An open standard process model that describes common approaches used by data mining experts.

References

- Runkler, T. A. (2020). Data analytics. Wiesbaden: Springer Fachmedien Wiesbaden.